Mohammadreza Nemati

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Experience

AI Engineer 05/2025 - Present
Volkswagen Group of America *Belmont, CA*

- Architected and deployed inference infrastructure for multimodal scene understanding and multi-agent LLM systems, designing evaluation frameworks with benchmarks for latency, throughput, and end-to-end execution accuracy.
- Fine-tuned multimodal foundation models (Qwen3-VL-8B) using SFT, DPO, and GRPO for post-training alignment, and applied Knowledge Distillation (KD) from 8B teacher to 2B student model with quantization-aware training to optimize inference efficiency.
- Implemented agentic RAG system managing dynamic UI tree hierarchy for agent-based task planning.
- Deployed and optimized production inference servers using vLLM and TensorRT, implementing continuous batching and profiling system bottlenecks to improve throughput and reduce serving latency at scale.

Artificial Intelligence Research Assistant 08/2022 - 06/2025
Case Western Reserve University *Cleveland, Ohio*

- Improved LLM KV-cache quantization by analyzing key/value norm imbalance, achieving 2× memory reduction with >94% accuracy retention across models up to 70B parameters, directly optimizing inference memory efficiency for large-scale serving [paper link]
- Developed LIT-LVM using latent variable models with structured regularization and GNN-based higher-order interactions for biomedical data, outperforming GNN, Factorization Machine, and elastic net baselines in ranking and recommendation tasks [paper link]
- Developed DEASurv, a deep survival model leveraging entity-specific embeddings, attention mechanisms, and contrastive loss to improve multi-modal biological survival prediction on large-scale medical datasets

Software Engineer - AI/ML Intern 02/2024 - 07/2024
Sherwin-Williams *Cleveland, Ohio*

- Designed and deployed real-time inference pipeline for vision transformers (SAM) with client-backend architecture, optimizing model latency and achieving 7% IoU improvement through inference optimization techniques.
- Built multimodal system integrating vision-language models (LLaVA) with vision transformers, implementing cross-modal alignment for natural language-driven visual grounding and object segmentation.

Data Science Research Intern 05/2023 - 09/2023
Beyond Limits AI *Glendale, California*

- Developed and deployed Graph Neural Network models (GCN, GAT, GraphSAGE) using PyTorch Geometric for graph-structured data, achieving 10% accuracy improvement over baseline approaches
- Fine-tuned Llama-2 LLM for domain-specific applications using distributed training (FSDP) for multi-GPU setups and quantization for efficient deployment.

Machine Learning Research Assistant 08/2018 - 07/2022
University of Toledo *Toledo, Ohio*

- Developed and deployed large-scale ML pipelines processing 500,000+ patient records with PyTorch, applying distributed data processing and statistical modeling to predict clinical outcomes

Skills

- **Programming Languages**: Python, C++
- **Machine Learning**: Multi-Modal Foundation Models, LLM/VLM Fine-tuning (SFT, DPO, GRPO), Knowledge Distillation, Vision Transformers, Reinforcement Learning, Graph Neural Networks
- **ML Infrastructure**: Distributed Inference Systems, Model Serving & Deployment, Distributed Training (FSDP), Performance Profiling & Optimization
- **Frameworks & Libraries**: PyTorch, vLLM, LangChain/LangGraph, PyTorch Geometric (PyG), NumPy, Pandas
- **Inference Optimization**: vLLM, TensorRT, CUDA, Quantization, Model Distillation, Continuous Batching
- **DevOps & Tools**: Docker, Flask, Redis, SQL, Git

Education

Case Western Reserve University Cleveland, Ohio
Doctorate (PhD), Computer Science *08/2022 - 05/2025*

University of Toledo Toledo, Ohio
Masters (MS), Computer Science *08/2018 - 08/2020*

Shiraz University Shiraz, Fars
Bachelors (BS), Electrical Engineering *08/2012 - 05/2017*